chCase Study : Hospital Management System

Objective :

1.To understand the role of software metrics in project estimation.
2.To apply LOC, Function Point, Productivity, and COCOMO techniques for effort and time calculation.
3.To estimate cost and resources required for a software project.

Step 1: Identify Metrics

Decide which metric to use for estimation:
1. Function Points (FP) – based on inputs, outputs, files, inquiries, interfaces.
2. LOC (Lines of Code) – derived from FP using the assumed productivity.
3. COCOMO Model – uses LOC + Salary per month.

Step 2: Collect Basic Data

For Hospital Management System

External Inputs (EI): Patient Registration, Doctor Registration, Add Room, Add Appointment, Add Bill, Add Test Report, Login **– 7**
External Outputs (EO):Patient Report, Bill/Invoice, Test Report, Appointment Report, Discharge Summary – **5**
External Inquiries (EQ):Search Patient, Check Doctor Availability, Check Room Availability, View Appointment Status – **4**
Internal Logical Files (ILF): Patient DB, Doctor DB, Room DB, Appointment DB, Billing DB, Test Report DB **– 6**
External Interface Files (EIF):payment system, External Laboratory system **– 2**

Step 3: Calculate Size (FP / LOC)

Use Average weights

Component	Simple	Average	Complex
External Input (EI)	3	4	6
External Output (EO)	4	5	7
External Inquiries (EI)	3	4	6
External Logical Files (ELF)	7	10	15
External Interface Files (EIF)	5	7	10

Unadjusted FP (UFP):
UFP=(7×4) + (5×5) + (4×4) + (6×10) + (2×7)
UFP=28 + 25 + 16 + 60 + 14 =**143**

Value Adjustment Factor (VAF): 0.65+(0.01×Total Degree of Influence (DI))
Where DI = sum of ratings (0–5) across 14 General System Characteristics

No.	GSC	Rating 0-5	Reason
1	Data communications	4	HMS communicates with database, email/SMS services, and payment systems.
2	Distributed data processing	3	Patient, doctor, billing, and laboratory data may be processed across different systems.
3	Performance	4	The system requires quick response for patient registration, appointments, and billing.
4	Heavily used configuration	3	Doctors, staff, receptionists, and administrators may use the system simultaneously.
5	Transaction rate	4	Frequent transactions include registrations, appointments, billing, and test reports.
6	Online data entry	5	Patient, doctor, appointment, billing, and test information is entered online.
7	End-user efficiency	4	Staff need an easy-to-use interface for quickly accessing patient and hospital records.
8	Online update	4	Patient records, appointments, room availability, and billing information are updated online.
9	Complex processing	3	Billing, appointment management, room allocation, and report generation involve moderate complexity.
10	Reusability	3	Common modules such as login, billing, and report generation can be reused.
11	Installation ease	3	Web-based HMS requires moderate installation and configuration effort.
12	Operational ease	4	The system should be easy for doctors, nurses, receptionists, and administrators to operate.
13	Multiple sites	3	The HMS can support multiple departments, wards, or hospital branches.
14	Facilitate change	4	Hospital services, doctors, departments, and billing rules may change, so the system should be flexible.

Total DI Calculation

DI = 4 + 3 + 4 + 3 + 4 + 5 + 4 + 4 + 3 + 3 + 3 + 4 + 3 + 4

DI = 51

Now calculate VAF:

VAF=0.65+(0.01×51)=0.65+0.51=**1.16**

Final FP (after VAF):

FP=UFP×VAF=143×1.16=**166 FP**

Language ratio**: Java ≈ 53 LOC/FP**

Estimated LOC:

LOC=166×53=**8,798 LOC(~~8.80KLOC)**

Step 4: Convert Size to Effort, Time & Cost

Constants:

Mode	a1	a2	b1	b2
Organic	2.1	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Effort =a1×(KLOC) ^a2

Effort = 3.0 × (8.798)^1.12 = 34.19 PM

Time (months)= b1×(Effort)^b2

Time = 2.5 × (34.19)^0.35 = 8.59 months

Team size = Effort/Development time

Team Size = 34.19 / 8.59 = 3.98 ≈ 4 persons

4. Cost:

Cost: If salary per PM = ₹50,000

Total Cost = 34.19 × ₹50,000 = ₹17,09,500 ≈ ₹17.1 Lakh

Step 5: Present Results

Estimated effort ≈ 34.2 PM, duration ≈ 8.6 months, team size ≈ 4 persons, and total cost ≈ ₹17.1 lakh for the Hospital Management System